

Project Proposal: Yacht Dice Game

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1 Project Title

Yacht: A Multi-player Strategic Dice Game in F#

2 Overview

This project is a GUI-based implementation of the classic dice game "Yacht" for two to four local players. Players take turns rolling five dice to complete specific combinations and record their scores in a scoring table. The game emphasizes strategic decision-making in dice retention and score allocation over 12 rounds.

3 Requirements

1. **Start Menu:** Upon execution, a pop-up screen displays the game title, a player count selector (2P to 4P), and a "Start" button.
2. **Player Selection:** Players can adjust the number of participants using arrow buttons next to the player count display.
3. **Main Game Layout:** Clicking "Start" initializes the main screen, featuring the Yacht scoring table on the left and five dice with a "power gauge" bar on the right.
4. **Scoring Table UI:**
 - a. The top-left corner displays the current turn progress (e.g., 1/12).
 - b. Column headers display player names (P1–P4 by default), which can be customized by clicking and typing.
 - c. Rows include: Aces, Deuces, Threes, Fours, Fives, Sixes, Subtotal, +35 Bonus, Choice, 4 of a Kind, Full House, S. Straight, L. Straight, Yacht, and Total.
5. **Dice Rolling Mechanism:**
 - a. Players roll dice by holding and releasing the Space bar.
 - b. A gauge bar fills while the Space bar is held; a higher gauge percentage increases the number of shuffle iterations for the dice animation.
 - c. Each player can roll the dice up to three times per turn.

6. **Dice Interaction:** After a roll, players can "fix" dice by left-clicking them. Fixed dice are highlighted with a red outline and remain unchanged during subsequent rolls in that turn. Fixed dice can be unfixed by left-clicking them.
7. **Scoring Logic:** After rolling, a player must select an empty entry in their column. The score is calculated as follows:
 - **Aces to Sixes:** Sum of dice showing the corresponding number.
 - **Choice:** Total sum of all five dice.
 - **4 of a Kind:** Sum of all dice if four or more are identical; otherwise 0.
 - **Full House:** Sum of all dice if there is a set of three and a set of two; otherwise 0.
 - **S. Straight:** 30 points if dice include 1-2-3-4-5; otherwise 0.
 - **L. Straight:** 30 points if dice include 2-3-4-5-6; otherwise 0.
 - **Yacht:** 50 points if all five dice are identical; otherwise 0.
8. **Turn Management:**
 - a. Turns proceed from left to right according to the player order.
 - b. **Visual Feedback:** The column of the player whose turn is currently in progress will be highlighted (e.g., glowing border or background color change) to clearly indicate the active player.
 - c. Once all players have finished their turn, the round counter increases.
9. **Game End and Bonus:** After 12 rounds, the final score is calculated. If the sum of Aces through Sixes is 63 or higher, a +35 bonus is awarded. The player with the highest total score wins.

4 Example Interaction

- **Initial State:** The program shows the start menu. Player selects "3P" and clicks "Start."
- **User Input:** Player 1 holds the Space bar until the gauge is 80% full, then releases it.
- **Game Logic:** The dice shuffle intensely (due to 80% power) and land on [2, 2, 2, 5, 6].
- **Updated State:** The first column (P1) glows to indicate it is Player 1's turn. Player 1 clicks the three '2's to fix them and rolls again to try for a "4 of a Kind" or "Yacht."
- **Special Events:** Player 1 completes a 'Yacht' on the last roll. Player 1 chooses the "Yacht" entry, 50 points are added to the table, and the highlight moves to the next player's column.